

HIGH PERFORMANCE COMPUTING

Assignment-2

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BATCH: 11

Section 2: Naïve Matrix Multiplication ($O(n^3)$ compute)

Aim

Measure serial performance of a triple-loop matrix multiplication implemented in pure Python, and profile computational hotspots.

Requirements

- Python 3.8+
- Standard library: time, random, cProfile, pstats, io, tracemalloc.

Code:

```
import random
import time
import cProfile
import pstats
import io
import tracemalloc

# -----
# Generate a random n x n matrix
# -----

def generate_matrix(n):
    return [[random.random() for _ in range(n)] for _ in range(n)]

# -----
```

```

# Naïve matrix multiplication ( $O(n^3)$ )

# -----

def matmul_naive(A, B):
    n = len(A)

    C = [[0.0 for _ in range(n)] for _ in range(n)]

    for i in range(n):
        for j in range(n):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]

    return C

# -----

# Measure execution time

# -----

def run_timing(n):
    A = generate_matrix(n)
    B = generate_matrix(n)
    start = time.perf_counter()
    matmul_naive(A, B)
    end = time.perf_counter()
    print("=== Execution Time ===")
    print(f"Matrix size : {n} x {n}")
    print(f"Time taken : {end - start:.4f} seconds\n")

# -----

# CPU profiling using cProfile

# -----

def run_cpu_profile(n):
    A = generate_matrix(n)
    B = generate_matrix(n)
    profiler = cProfile.Profile()
    profiler.enable()
    matmul_naive(A, B)

```

```

profiler.disable()

s = io.StringIO()

stats = pstats.Stats(profiler, stream=s).sort_stats("cumulative")

stats.print_stats(10)

print("=== CPU Profiling (Top 10) ===")

print(s.getvalue())


# -----

# Memory profiling using tracemalloc

# -----

def run_memory_profile(n):

    A = generate_matrix(n)

    B = generate_matrix(n)

    tracemalloc.start()

    matmul_naive(A, B)

    current, peak = tracemalloc.get_traced_memory()

    tracemalloc.stop()

    print("=== Memory Profiling ===")

    print(f"Current memory usage : {current / 10**6:.2f} MB")

    print(f"Peak memory usage : {peak / 10**6:.2f} MB\n")

# -----

# Main function

# -----

def main():

    N = 200 # Change carefully: 100, 200, 300 (higher = much slower)

    print("\nNaïve Matrix Multiplication Performance Analysis\n")

    run_timing(N)

    run_cpu_profile(N)

    run_memory_profile(N)

if __name__ == "__main__":

    main()

```

LOCAL OUTPUT:

```
Naive Matrix Multiplication Performance Analysis

=== Execution Time ===
Matrix size : 200 x 200
Time taken : 0.6044 seconds

=== CPU Profiling (Top 10) ===
      4 function calls in 0.636 seconds

Ordered by: cumulative time

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
   1    0.636    0.636    0.636    0.636 C:\Users\ARUN REDDY\Downloads\Ladder-Problem.py:15(matmul_naive)
   1    0.000    0.000    0.001    0.001 C:\Users\ARUN REDDY\Downloads\Ladder-Problem.py:17(<listcomp>)
   1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
   1    0.000    0.000    0.000    0.000 {built-in method builtins.len}

=== Memory Profiling ===
Current memory usage : 0.01 MB
Peak memory usage   : 1.29 MB

>>>
```

GOOGLE COLAB

CPU:

```
... Naive Matrix Multiplication Performance Analysis

=== Execution Time ===
Matrix size : 200 x 200
Time taken : 1.3545 seconds

=== CPU Profiling (Top 10) ===
      355 function calls (353 primitive calls) in 1.146 seconds

Ordered by: cumulative time
List reduced from 109 to 10 due to restriction <10>

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
   2    0.000    0.000    1.140    0.570 /usr/lib/python3.12/asyncio/base_events.py:1922(_run_once)
   2    0.000    0.000    1.129    0.565 /usr/lib/python3.12/selectors.py:451(select)
   2    0.484    0.242    1.129    0.565 {method 'poll' of 'select.epoll' objects}
   1    0.645    0.645    0.645    0.645 /tmp/ipython-input-4270066419.py:15(matmul_naive)
   2    0.000    0.000    0.017    0.008 /usr/lib/python3.12/asyncio/events.py:86(_run)
   2    0.000    0.000    0.017    0.008 {method 'run' of '_contextvars.Context' objects}
   8    0.016    0.002    0.016    0.002 /usr/local/lib/python3.12/dist-packages/zmq/sugar/socket.py:632(send)
   1    0.000    0.000    0.011    0.011 /usr/local/lib/python3.12/dist-packages/tornado/platform/asyncio.py:206(_handle_events)
   1    0.000    0.000    0.011    0.011 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:574(_handle_events)
   1    0.000    0.000    0.011    0.011 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:615(_handle_recv)

=== Memory Profiling ===
Current memory usage : 0.01 MB
Peak memory usage   : 1.29 MB
```

TPU:

```
... Naive Matrix Multiplication Performance Analysis

=== Execution Time ===
Matrix size : 200 x 200
Time taken : 0.5977 seconds

=== CPU Profiling (Top 10) ===
      490 function calls (486 primitive calls) in 0.540 seconds

Ordered by: cumulative time
List reduced from 123 to 10 due to restriction <10>

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
   3    0.000    0.000    0.540    0.180 /usr/lib/python3.12/asyncio/base_events.py:1922(_run_once)
   3    0.000    0.000    0.524    0.175 /usr/lib/python3.12/selectors.py:451(select)
   1    0.318    0.318    0.318    0.318 /tmp/ipython-input-4270066419.py:15(matmul_naive)
   2    0.205    0.103    0.205    0.103 {method 'poll' of 'select.epoll' objects}
   3    0.000    0.000    0.016    0.005 /usr/lib/python3.12/asyncio/events.py:86(_run)
   3    0.000    0.000    0.016    0.005 {method 'run' of '_contextvars.Context' objects}
   8    0.015    0.002    0.015    0.002 /usr/local/lib/python3.12/dist-packages/zmq/sugar/socket.py:632(send)
   2    0.000    0.000    0.011    0.005 /usr/local/lib/python3.12/dist-packages/tornado/platform/asyncio.py:206(_handle_events)
   2    0.000    0.000    0.011    0.005 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:574(_handle_events)
   2    0.000    0.000    0.010    0.005 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:615(_handle_recv)

=== Memory Profiling ===
Current memory usage : 0.01 MB
Peak memory usage   : 1.29 MB
```

GPU:

```
Naive Matrix Multiplication Performance Analysis

*** === Execution Time ===
Matrix size : 200 x 200
Time taken  : 0.8500 seconds

=== CPU Profiling (Top 10) ===
491 function calls (486 primitive calls) in 0.778 seconds

Ordered by: cumulative time
List reduced from 123 to 10 due to restriction <10>

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
4/3     0.000    0.000    0.777    0.259 /usr/lib/python3.12/asyncio/base_events.py:1922(_run_once)
3       0.001    0.000    0.761    0.254 /usr/lib/python3.12/selectors.py:451(select)
1       0.555    0.555    0.555    0.555 /tmp/ipython-input-4270066419.py:15(matmul_naive)
2       0.205    0.103    0.205    0.103 {method 'poll' of 'select.epoll' objects}
3       0.000    0.000    0.016    0.005 /usr/lib/python3.12/asyncio/events.py:86(_run)
3       0.000    0.000    0.016    0.005 {method 'run' of 'contextvars.Context' objects}
2       0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/tornado/platform/asyncio.py:206(_handle_events)
2       0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:574(_handle_events)
2       0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:615(_handle_rcv)
2       0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:547(_run_callback)

=== Memory Profiling ===
Current memory usage : 0.01 MB
Peak memory usage   : 1.29 MB
```